The Incidence and Real Activity Effects of Commercial Property Taxes: Firm-Level Evidence

Nathan Sunday

March 27, 2026

Motivation and Research Question

- Property taxation is widely viewed as holding significant untapped revenue potential in developing countries.
  - Countries can to raise substantial revenue by increasing compliance from currently low levels.
  - Only about 10% pay annual liabilities (e.g., Manwaring and Regan 2025).
- Recent work studies policies and administrative reforms to improve property tax compliance and revenue (e.g., Manwaring and Regan 2025; Flores et al. 2025; Bergeron et al. 2024).

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- **However:** Evidence on the efficiency cost of property taxation and its distributional consequences remains limited.
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- **Questions:** How do commercial property taxes affect real economic activity? Who bears the burden of the tax?

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- Study a reform that increased property tax liabilities in Kampala city but not the surrounding areas.
- Use staggered DID to estimate causal effects on business rental expenses, real activity, and incidence on workers.

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- Use staggered DID to estimate causal effects on business rental expenses, real activity, and incidence on workers.
- **Results**
  - The reform increased business rental expenses by 9%, citywide pass-through rate of 0.65.
  - No significant effects on firms revenue but profits declined by 3.3 p.p while investment rate declined by 2.7%.
  - Reform reduced formal employment by 4.5% and firm survival probability by 3.2%.
  - Workers at the top of the earnings distribution were disproportionately affected.

Literature and Contribution

- Effects of local taxes on business decisions
  - Effects state and local tax policies on business location choices (Rohlin et al.2014; Fajgelbaum et al. 2019; Giroud and Rauh 2019) and employment decisions (Thompson and Rohlin 2012; Giroud and Rauh 2019).
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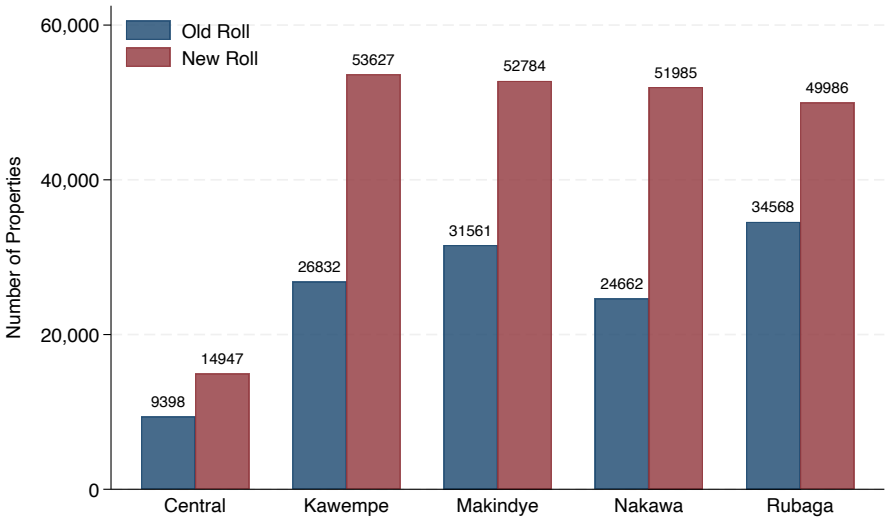
  - **Contribution:** This literature focuses on sales tax. I provide evidence on how property taxation affect businesses and its incidence on workers.
- Property tax incidence: England (2016) and Oates and Fischel (2016) for review
  - Recent work: Watson and Ziv (2024); Baker, (2024) document evidence of pass-through to rent costs in the case of rental housing.
  - **Contribution:** Provide evidence of pass-through in the context of commercial space and propose a framework to explain this result.

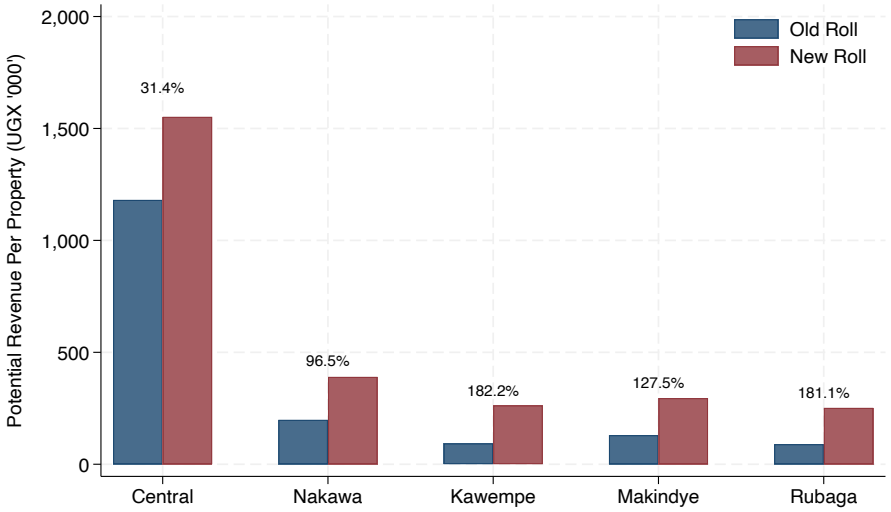
The Reform

- Property revaluation and enumeration in Kampala, following a ten-year hiatus.
- Increased the number of properties on the tax roll by 176%, the average property tax liability increased by $\approx$ 90%.

(a) Number of Properties



(b) Potential Revenue Per Property



Local Newspaper: May 23, 2017

Africa Edition Kenya Edition Uganda Edition Tanzania Edition

MONITOR

National Education Insight World

PRIME

KCCA increase property tax rates, landlords threaten to close shops

Tuesday, May 23, 2017 — updated on January 15, 2021

Rent. Tenants on Zai Plaza demonstrate recently over increased rent. The increase of property tax by KCCA will force landlords to hike rent charges for their structures. PHOTO BY RACHEL MABALA

What you need to know:

New rates. A landlord, who has been paying Shs15m property tax for a building, will now pay Shs45m by June 1, 2017.

Kampala. Kampala Capital City Authority (KCCA) has increased the amount of money by three times what it charges in property tax, a move that could lead landlords passing on the burden to tenants through rent.

Data and Sample Construction

1. Data
   - Firm-level panel data:
     - Universe of CIT firms in Uganda, 2014/15 - 2021/22.
     - Balance sheet, profit & loss accounts, tax information, industry, and location.
     - New data, first became available in 2021. Thanks to URA and UNU-WIDER.
   - Monthly employee tax returns. A universe of formal employees, linked to firms.

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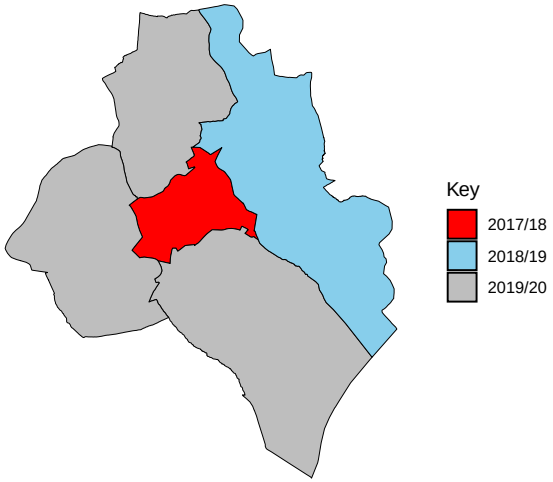
2. Sample Construction
   - Full CIT returns dataset comprises 299,683 firm–year observations from firms that are active at some point.
   - I restrict the sample to firms located in the Greater Kampala Metropolitan Area (GKMA); 213,185 firm–year observations.
   - Restrict to firms that had positive revenues in pre-reform year; 16,358 unique firms.
   - Additional restricts based on the sections.

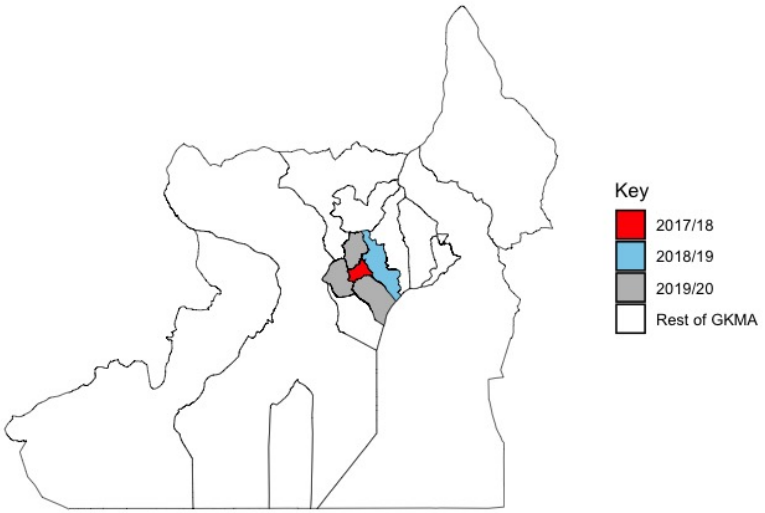
Identification

- Staggered DiD: Callaway & Sant'Anna (2021). CS DiD
- Firms outside Kampala, but within the Greater Kampala Metropolitan Area are the never treated units.
- *Identifying Assumption:* Absent the reform, average outcomes for treated and untreated firms would have evolved similarly.

(a) Kampala Capital City



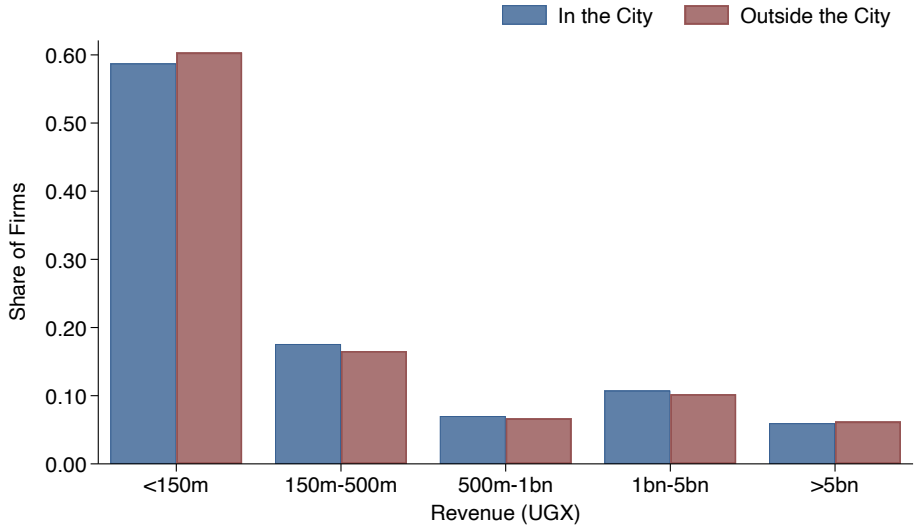
(b) Greater Kampala Metropolitan Area



Sample Characteristics

- There is substantial overlap along firms' revenues and formal employment distributions across treated and comparison locations.

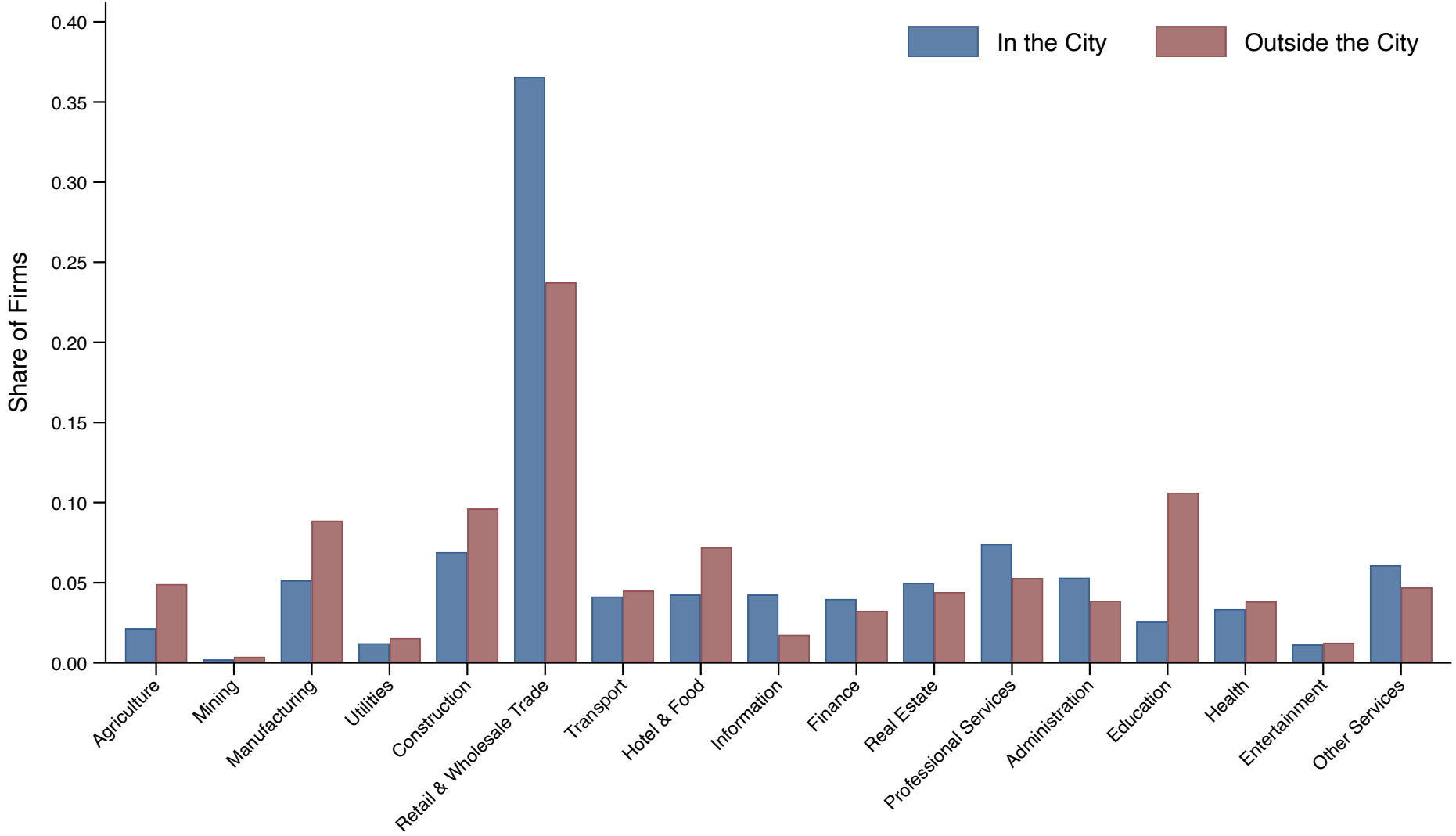
(a) Revenues (UGX)



(b) Formal Employees

Density

Sample Characteristics: Industry Composition



- More sample characteristics Table

Effects on Business Rental Expenses

- Does the burden entirely fall on landlords? — the neoclassical view.
- Outcome variable: Rental expenses normalized pre-reform values

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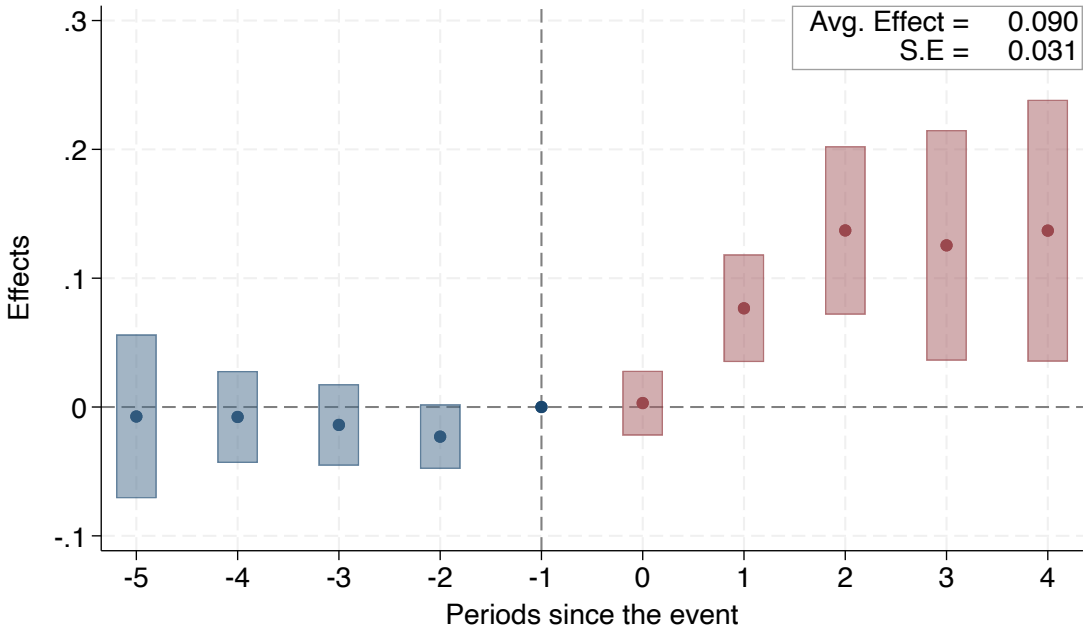

- Result: Increase in business rental expenses, citywide pass-through rate of 0.65.

Revenues and Profits

- Property taxes raise occupancy-related costs, both directly and indirectly through rent pass-through.
- Firms may adjust along different margins, affecting their performance.
- Sample: Balanced panel of active firms.

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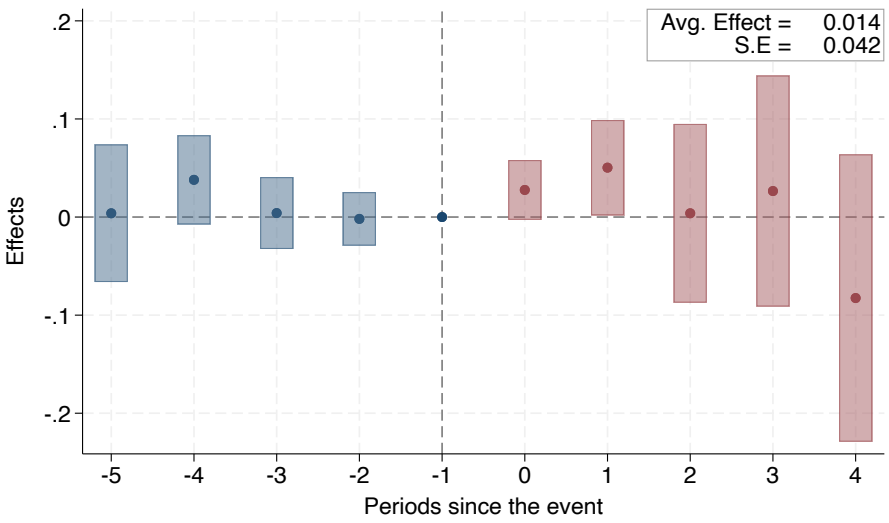
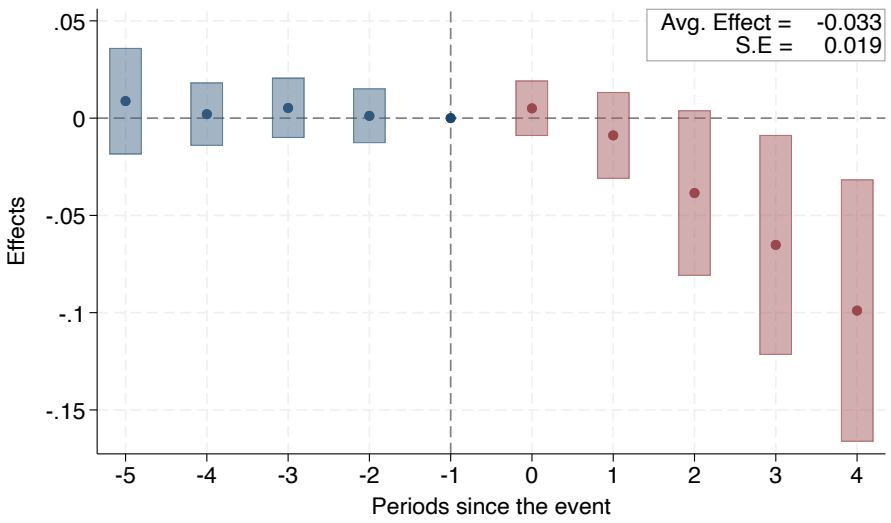
(a) Log Revenues

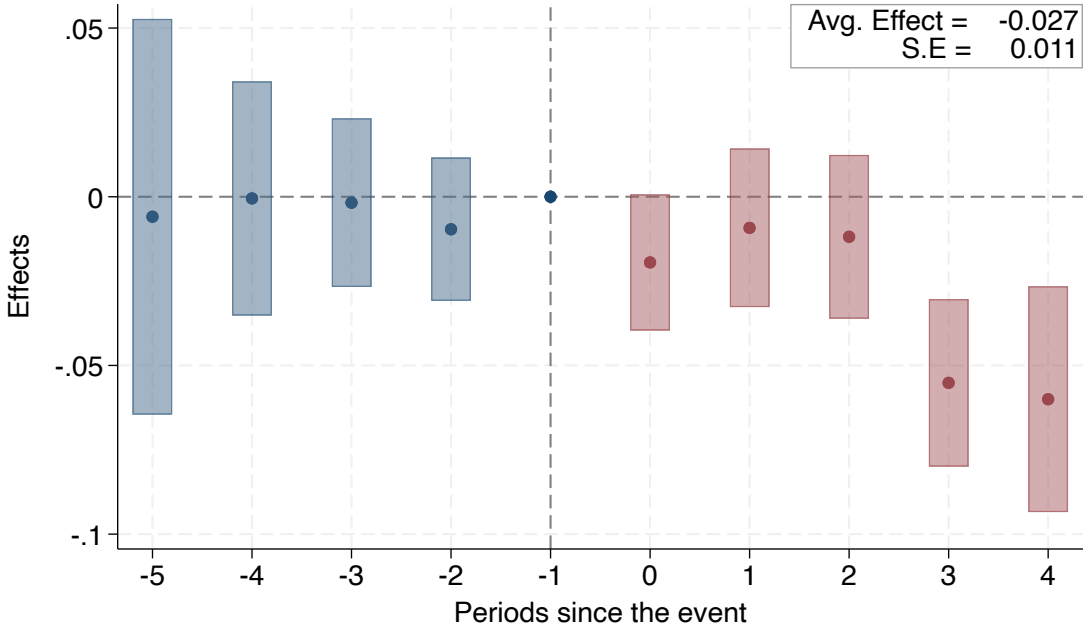


(b) Profits/Pre-reform Turnover



Effects on Investment

- Investment = Annual change in net fixed assets, scaled by lagged capital.



- Result: Decrease in investment rate by 2.7 pp.

Extensive Margin + Capital

Effects on Formal Employment

- Outcome: Average employment over the twelve months; from employee tax returns.
- Sample: Firms with positive revenues and non-zero employment in pre-reform.
- Normalize the outcome by pre-reform employment

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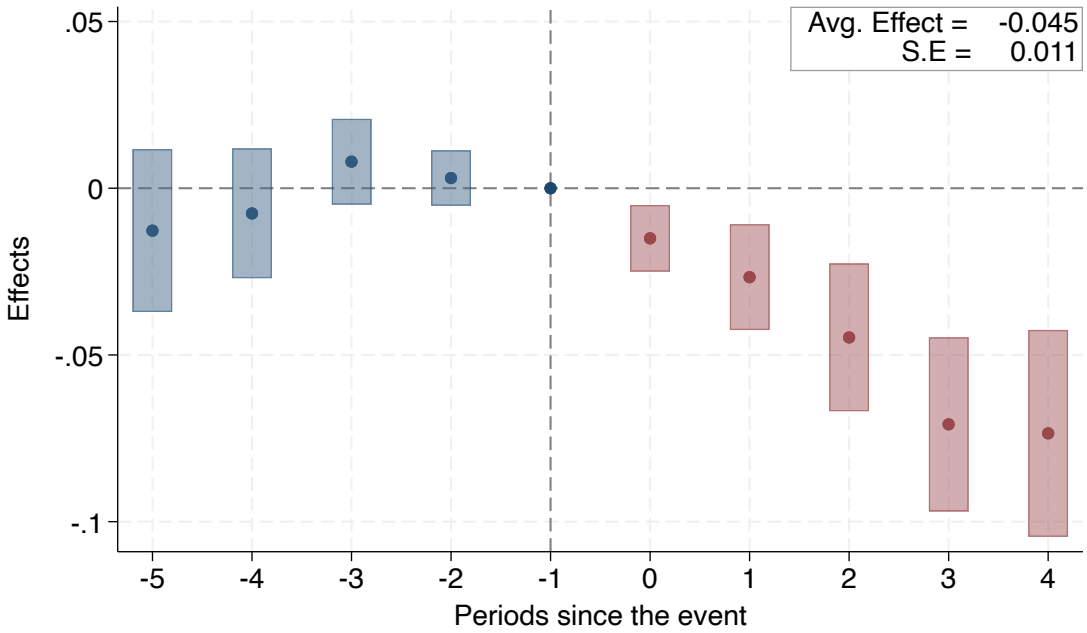


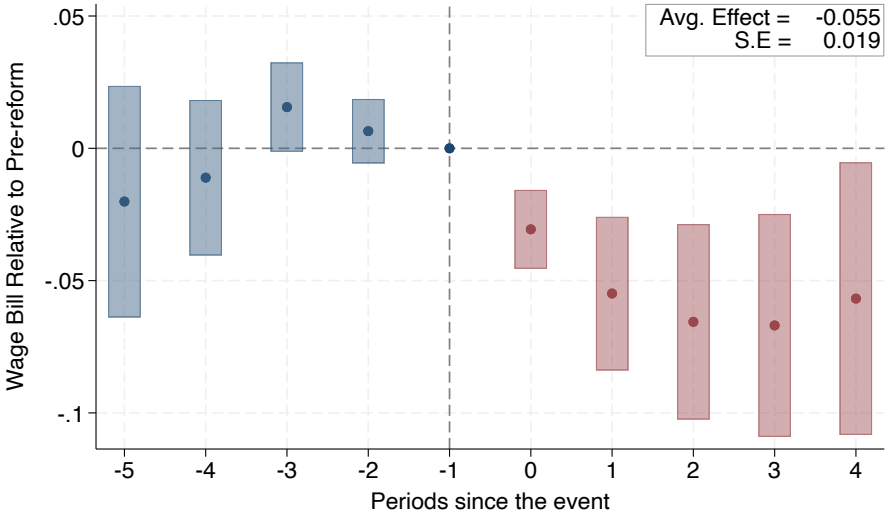
Logs; IHS; Levels

Balanced panel of active firms

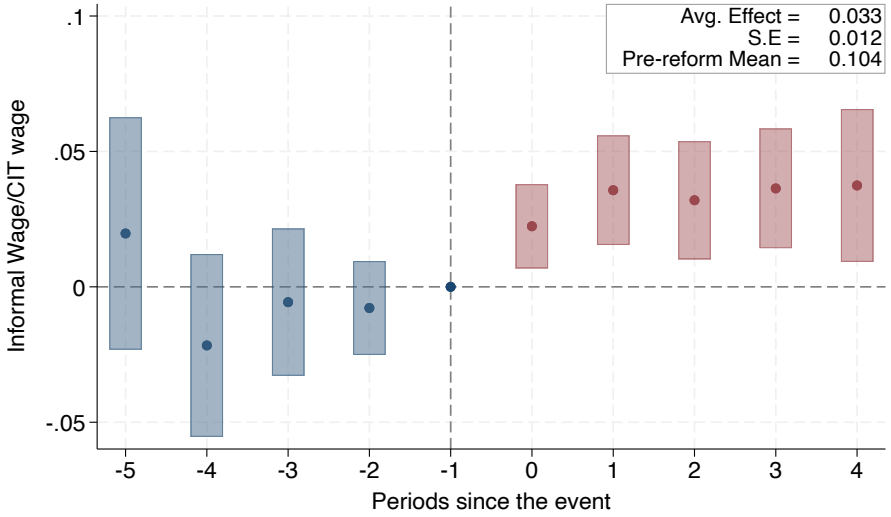
Effects on Firms' Wage Bill

- Formal wage bill reduces mechanically due to decline in employment.
- Informal wages (CIT wage bill - Formal wage bill) increase!
- Some of the formal jobs are becoming informal?

(a) Formal Wage Bill/Pre-reform Values



(b) Informal Wage Bill/CIT wage Bill



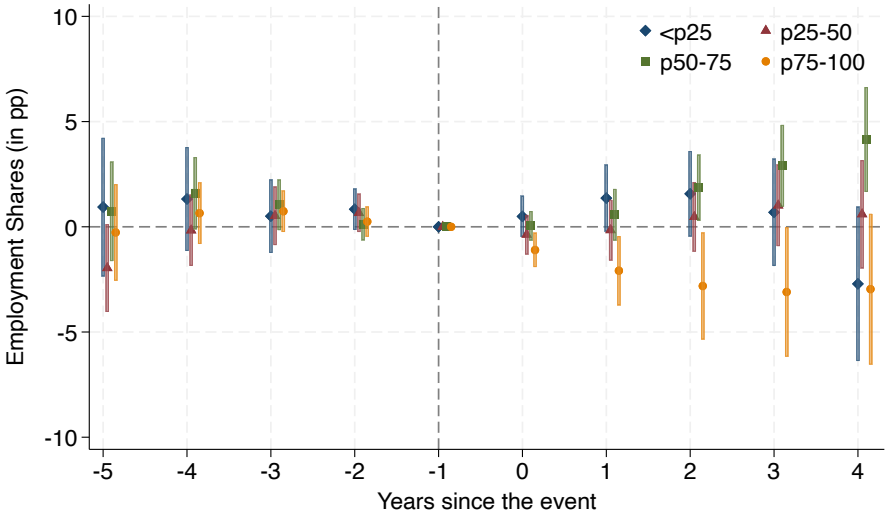
Firm-Level Distributional Consequences

- Aggregate employment effects mask important distributional consequences.
- Moreover, firms may adjust wages or worker composition without necessarily changing total employment.
- **Goal of this part:** Which workers are most affected?
- Data limitations preclude worker-level analysis
  - Workers are uniquely identified only at the firm level. details
- Conduct firm-level analysis: study the evolution of the shares of workers in each pre-reform wage quartile.
- Sample restriction: firms with at least five pre-reform workers, and remained active throughout the sample period.
- Complementary analysis: study how the within-firm wage distribution changes.

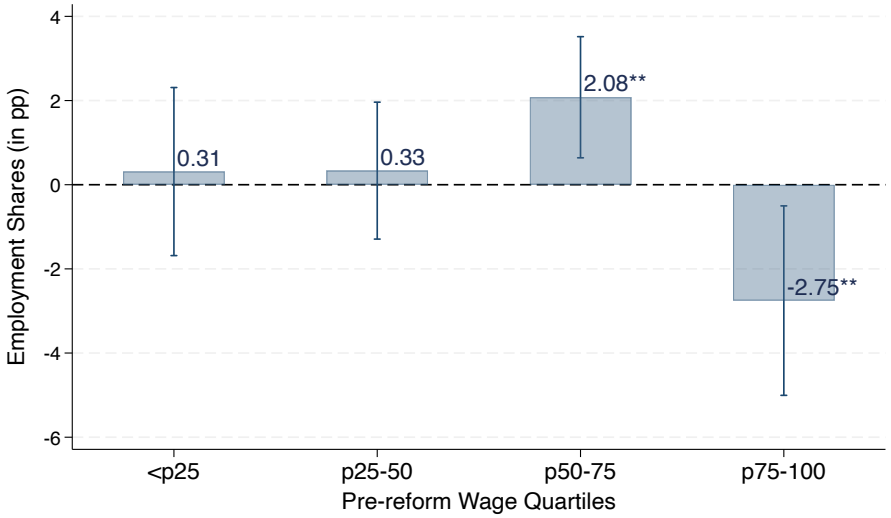
Effects on Worker Composition

- *Outcome variable:* Share of workers, relative to pre-reform employment, in each (fixed) bin.
- Differential decline in the share of workers in the top quartile.
- Differential increase in the share of workers in the 50th - 75th percentile.

(a) Event-study



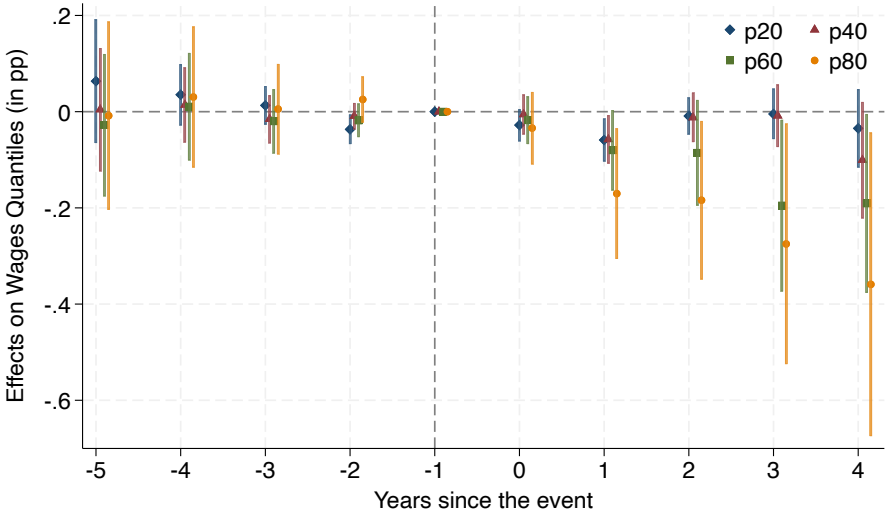
(b) Average Effect



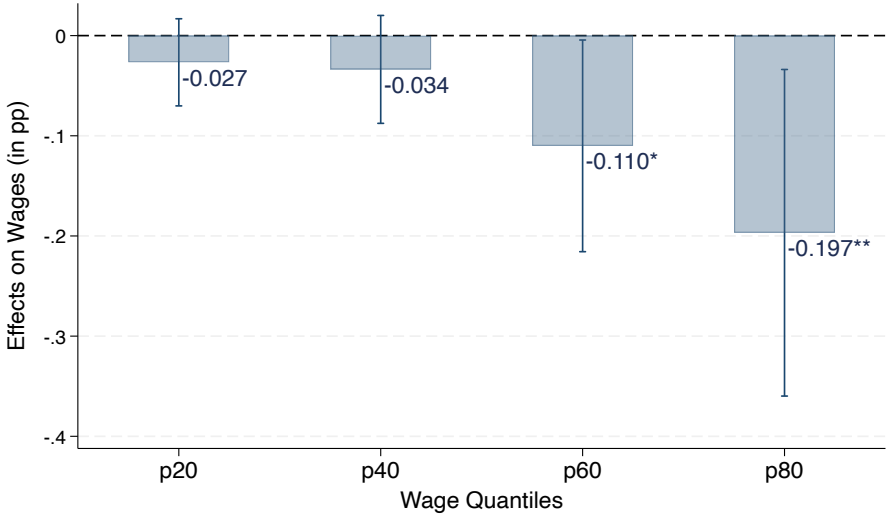
Effects on the Within-firm Wage Distribution

- One potential explanation for the previous pattern is wages at the top of the distribution declined more than those at the bottom.
- *Outcome variable:* Wage quantiles, normalized by pre-reform wage bill.

(a) Event-study

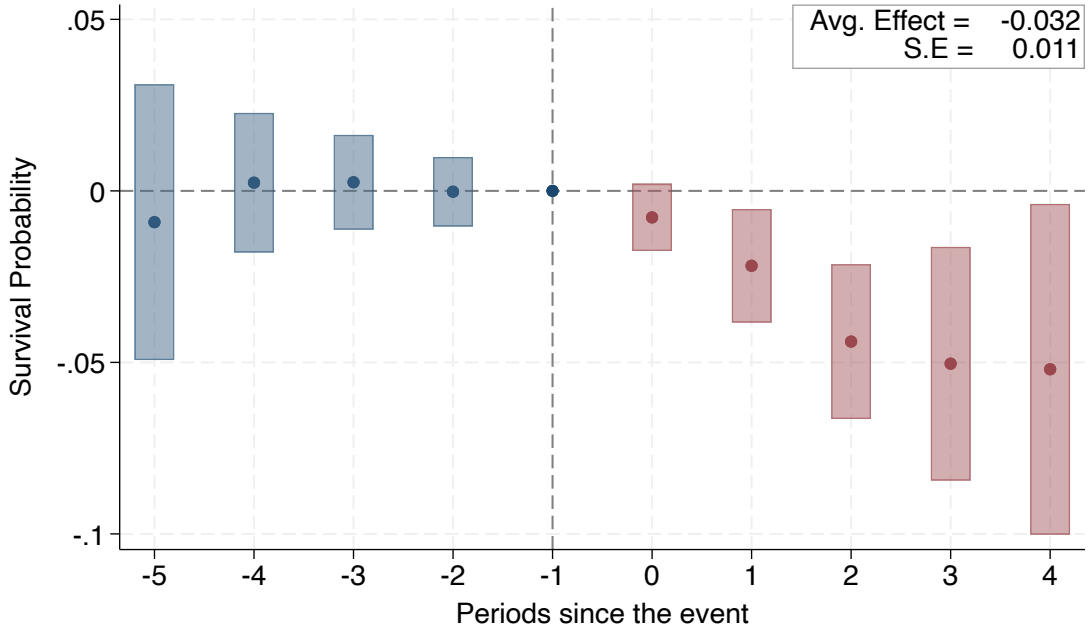


(b) Average Effect



Effects on Firm Survival

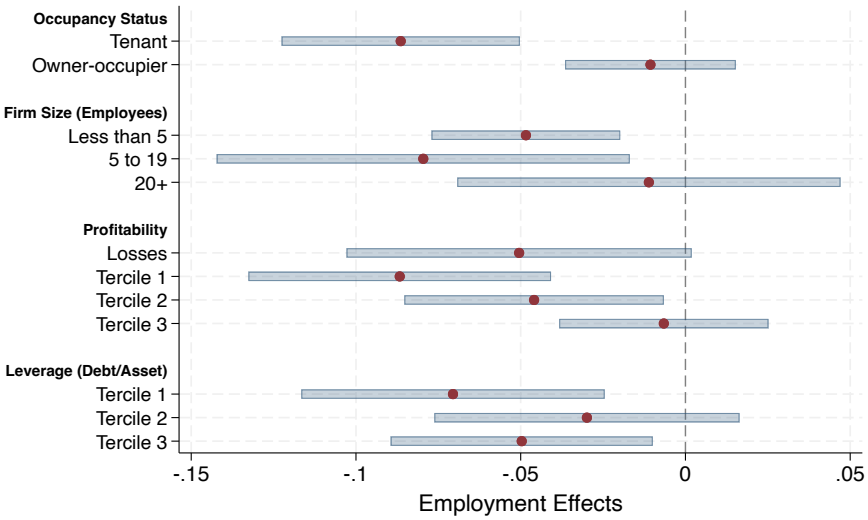
- Sample: Firms present in 2015/16 (pre-reform) with non-zero turnover, regardless of activity in other years (similar to Saez et al. (2019)).
- Robustness: Restrict to firms present in the first panel year.
- Set turnover $= 0$ in years with no operation.
  $\text{Survival}_t = \mathbb{I}\{\text{turnover}_t > 0\}$.



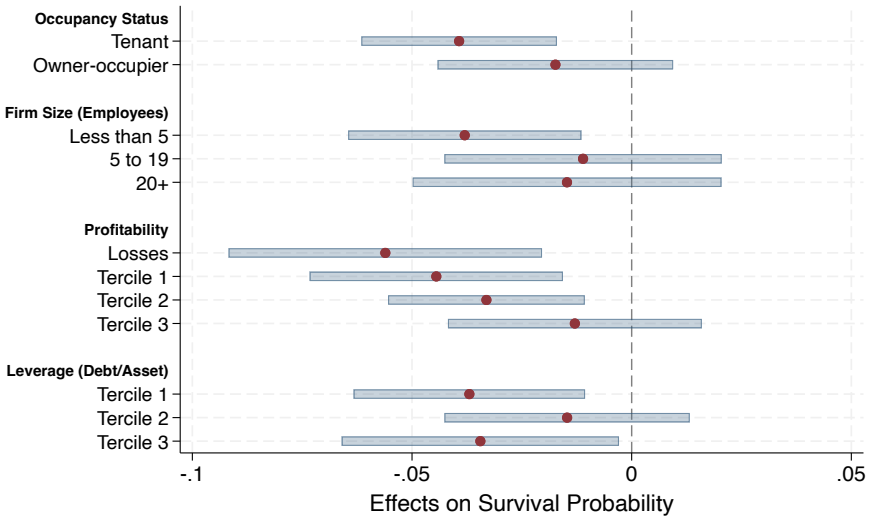
Survival or Evasion? Robustness at t and $t+1$ First year of the sample

Heterogeneity by Firm Characteristics

(a) Employment Effects



(b) Survival Effects



- Effects are higher among less profitable firms.
- Firms with very low and very high leverage are more affected.

Summary and Conclusion

- The paper presents three key finding:
  1. A substantial share of the burden of commercial property taxes falls on tenants, contrary to the neoclassical prediction.
  2. Commercial property taxation has important implications for real activity: negatively affects firms' profitability, investment, employment and survival.
  3. Within firm, the burden was not borne equally by all workers.
     - Workers at the top of the earnings distribution were disproportionately affected.
     - The reform reduced within-firm wage inequality and was therefore progressive.
     - This pattern is also consistent with skill downgrading within firms.
- Profitability and leverage play a role in explaining the results.

Comments: nsunday@umich.edu

Identification Strategy

- TWFE regression:

$$y_{it} = \alpha_g + \alpha_t + \sum_{\substack{e=-5 \\ e\neq -1}}^{4} \beta_e D_{it}^e + \gamma X_{it} + \nu_{it},$$

- Staggered Difference-in-Difference: Callaway & Sant'Anna (2021).

$$\beta_{es}(e) = \sum_{g\in\mathcal{G}} \mathbb{I}\{g+e \leq \mathcal{T}\}\, \mathbb{P}(G=g \mid G+e \leq \mathcal{T})\, ATT(g, g+e)$$

  - $e = t - g$ is length of exposure.
  - $ATT(g,t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G_g = 1]$; Average treatment effect for units who are members of a particular group g at a particular time period t
  - $\beta_{es}(e)$: Average across all groups that are ever observed to have participated in the treatment for exactly e time periods.

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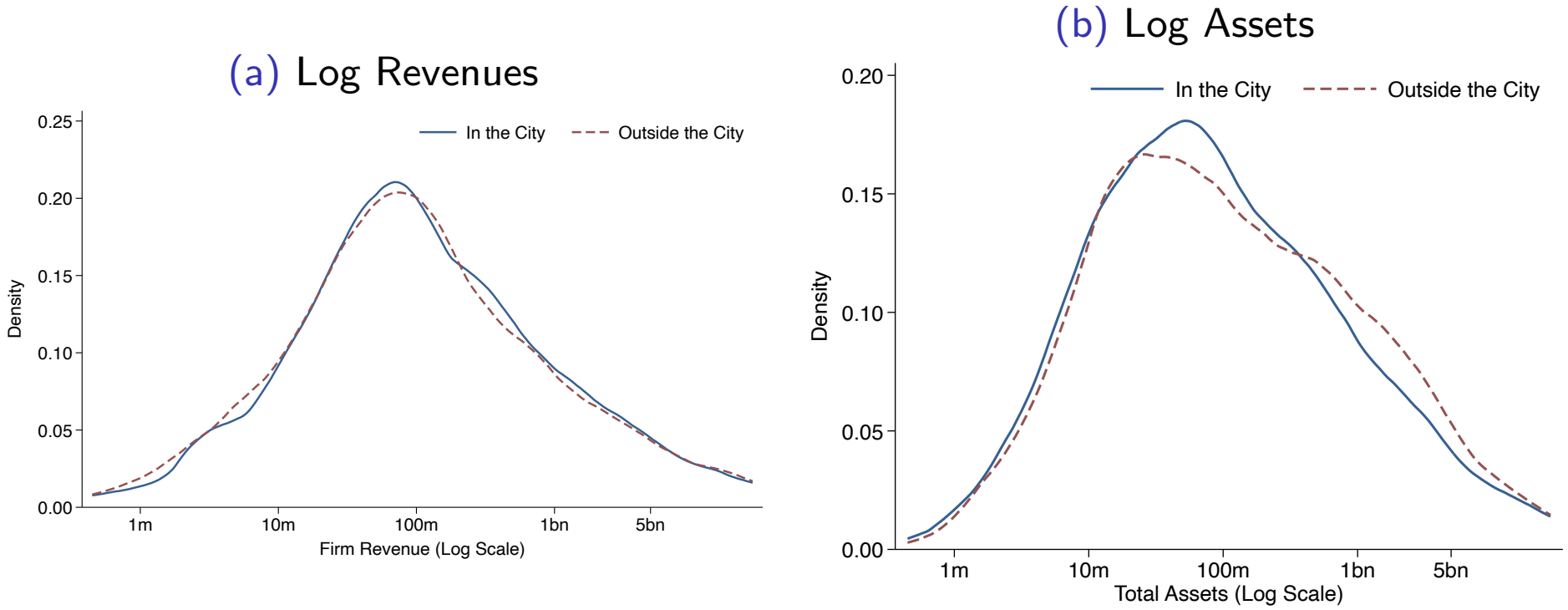
Sample Characteristics: Density



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Sample Characteristics

	(1) Universe	(2) Ever-Treated	(3) Comparison Group
Revenues	556.09	614.96	629.81
	(1,232.30)	(1,294.14)	(1,328.00)
Total Assets	267.43	284.87	361.35
	(595.70)	(620.41)	(696.65)
Fixed Assets	188.27	190.70	294.15
	(461.14)	(468.84)	(568.66)
Formal Employees	5.16	5.15	7.45
	(10.69)	(10.43)	(13.00)
Rent Expenses	6.15	8.39	4.43
	(12.84)	(14.83)	(11.27)
Landing and Buildings	72.33	63.74	130.18
	(209.02)	(202.33)	(270.43)
$\mathbb{I}\{\text{Rent Expense} > 0\}$	0.44	0.50	0.34
	(0.50)	(0.50)	(0.47)
$\mathbb{I}\{\text{Landing and Buildings} > 0\}$	0.20	0.15	0.31
	(0.40)	(0.36)	(0.46)
$\mathbb{I}\{\text{Loss Making}\}$	0.20	0.23	0.23
	(0.40)	(0.42)	(0.42)
Profit Margins	0.15	0.13	0.14
	(0.31)	(0.31)	(0.33)
Cash-to-Assets	0.17	0.17	0.13
	(0.22)	(0.23)	(0.20)
Debt-to-Assets	0.41	0.35	0.36
	(0.36)	(0.35)	(0.35)

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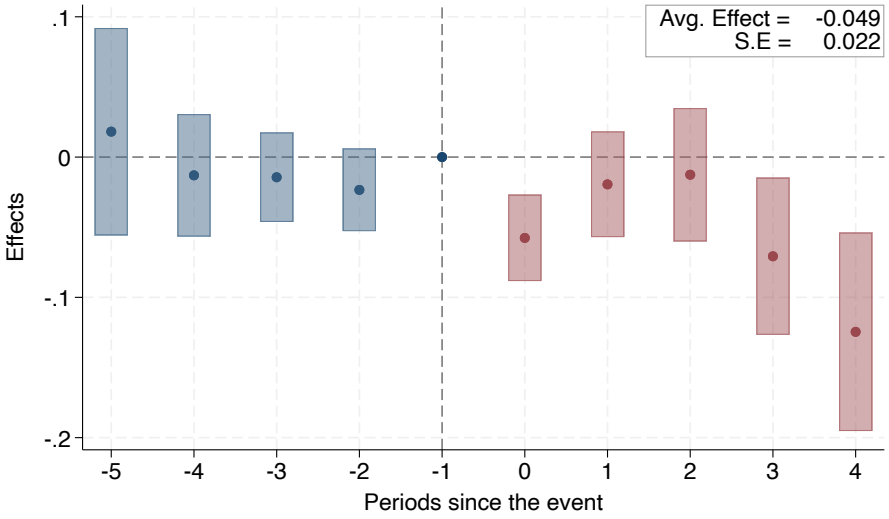
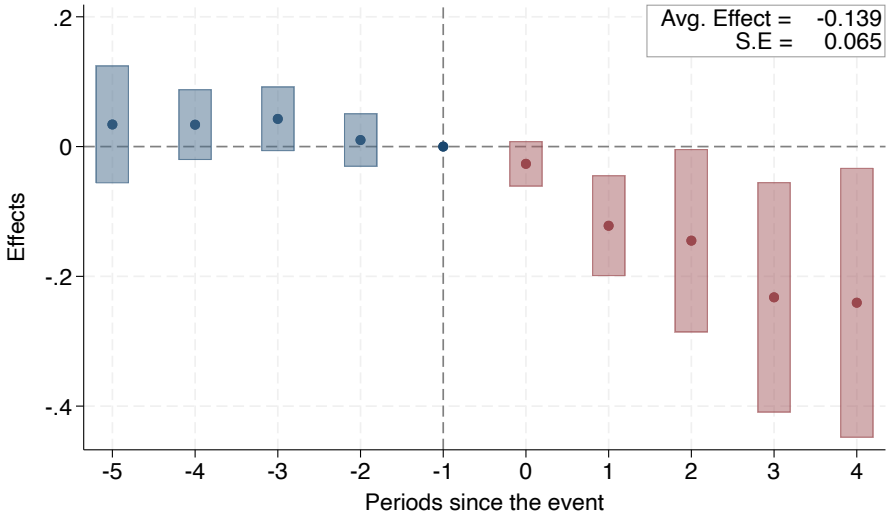
Investment at Extensive Margin + Capital Stock

(a) Investment: Extensive Margin



(b) Log Capital



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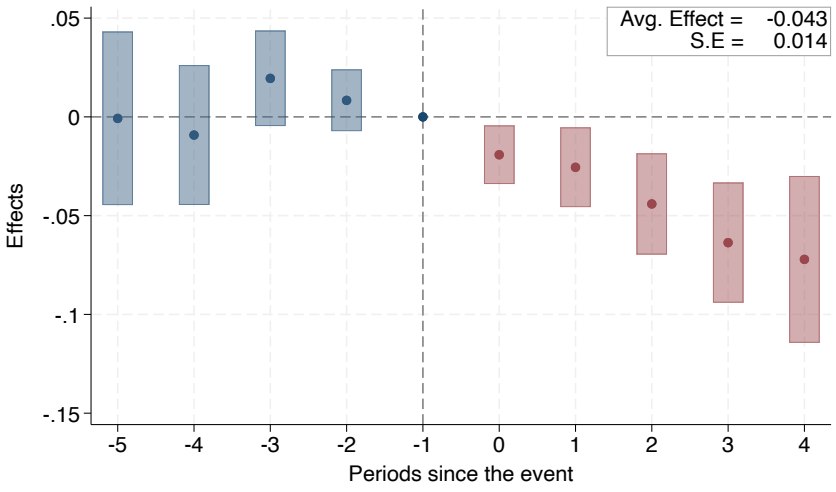
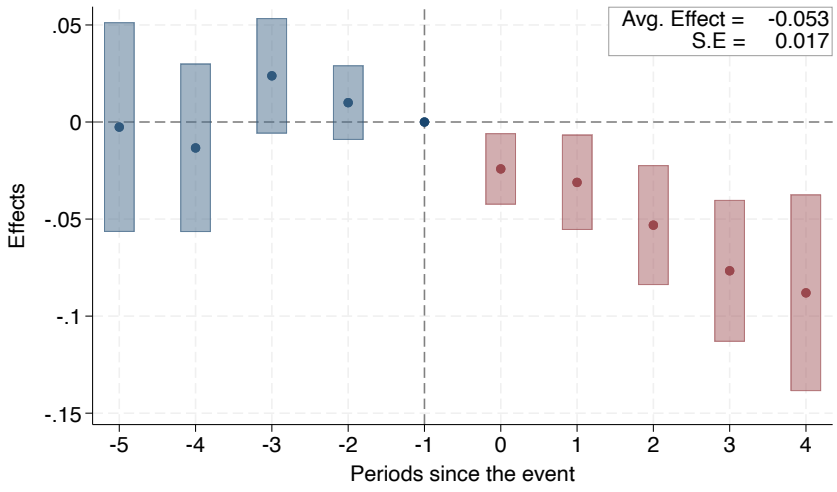
Effects on Formal Employment: Logs, IHS, and Levels

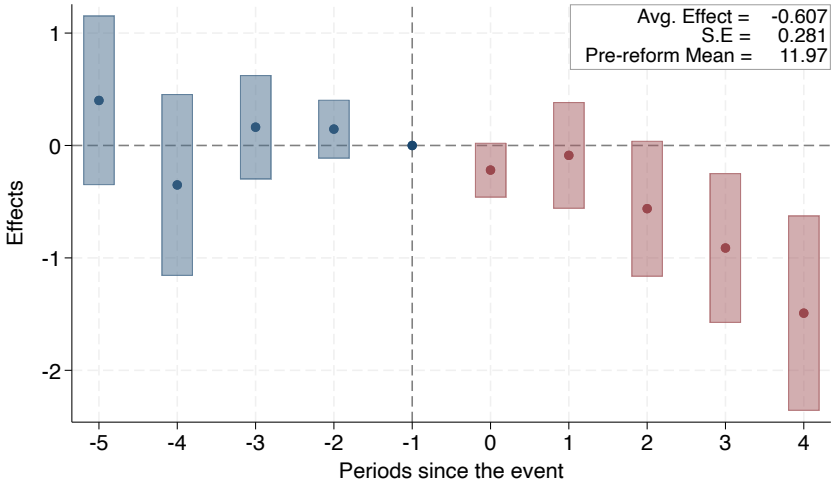
(a) Log (1+Employment)



(b) IHS(Employment)



(c) Employment in Levels



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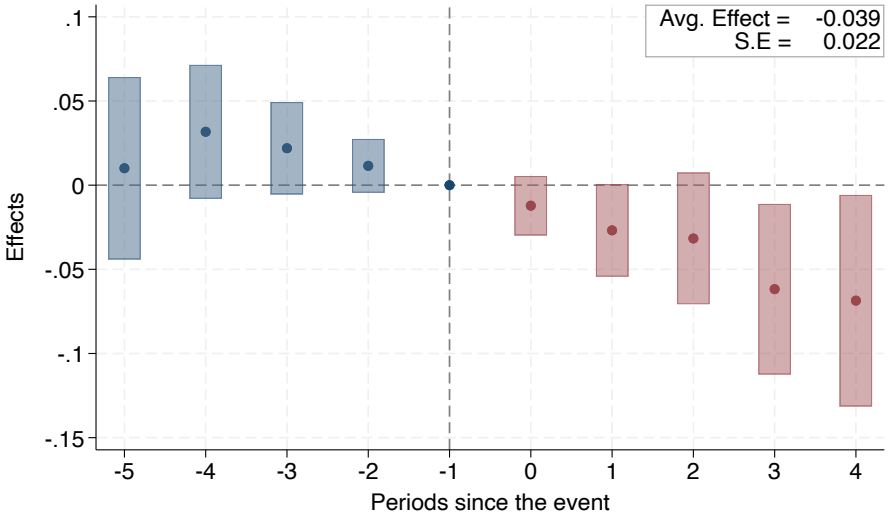
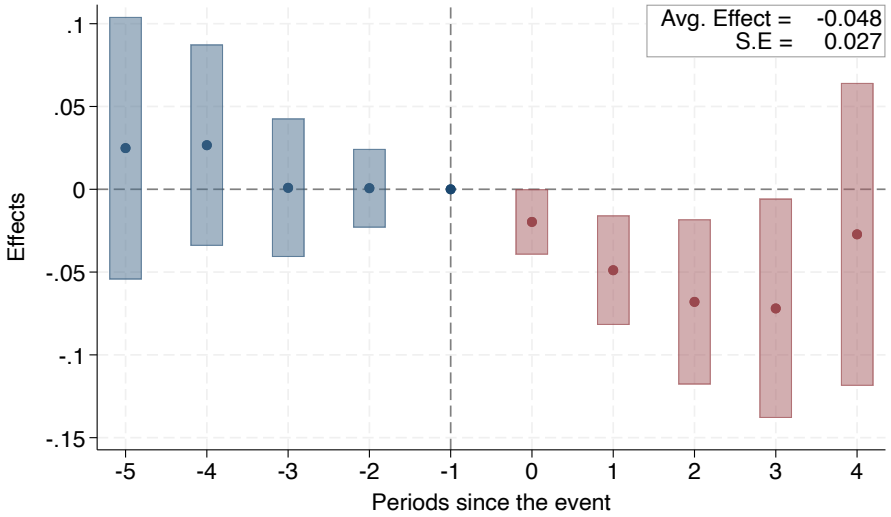
Effects on Formal Employment: Balanced Panel of Active Firms

(a) Active with Non-Zero Revenues



(b) Active with Non-Zero Employment



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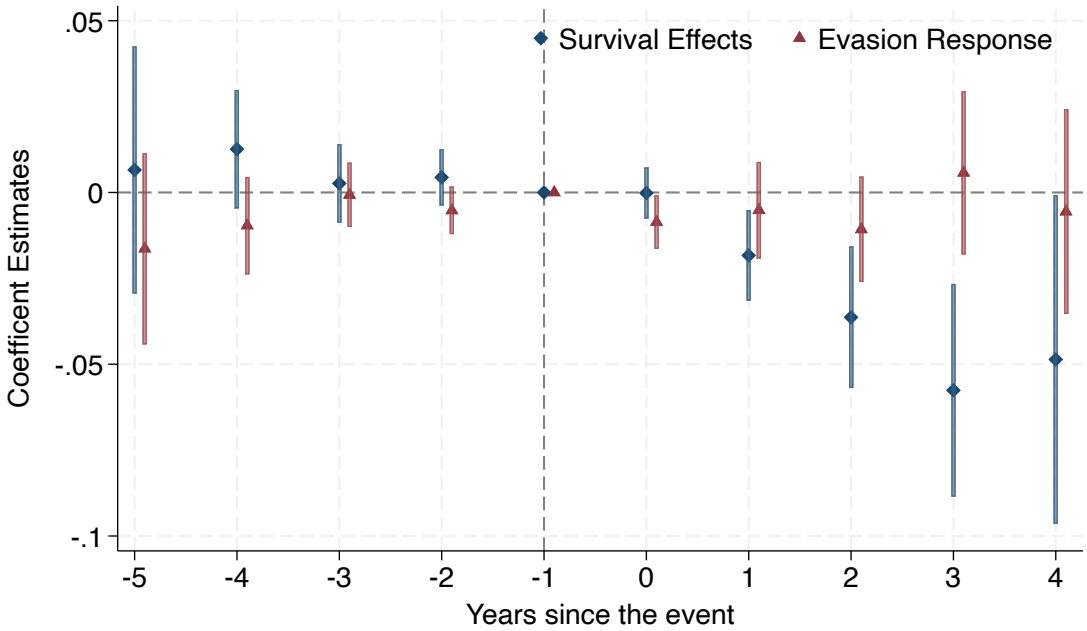
Evasion or Survival?

- Using cross-tax validation: CIT and employee tax returns.
  - Classify exit only if *no CIT return* and *no PAYE footprint*.
  - Evasion if *no CIT* return but has footprints in employee returns.
- Effects not driven by evasion response.



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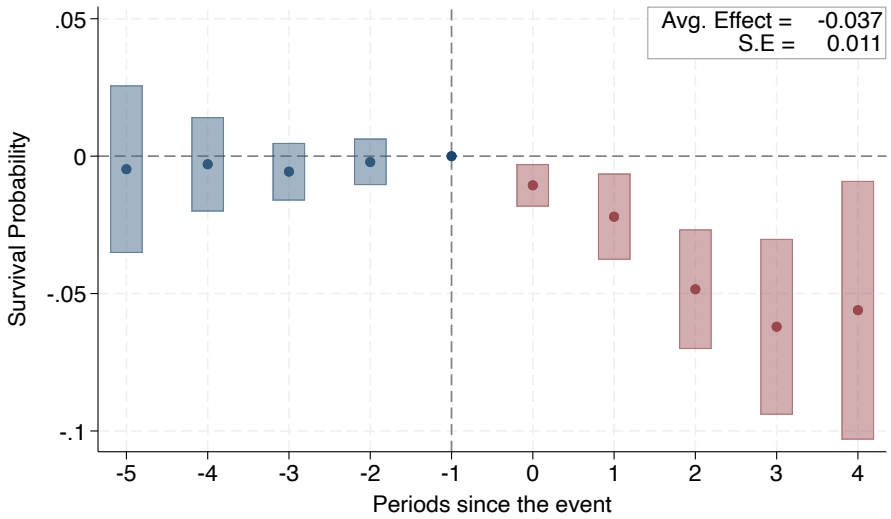
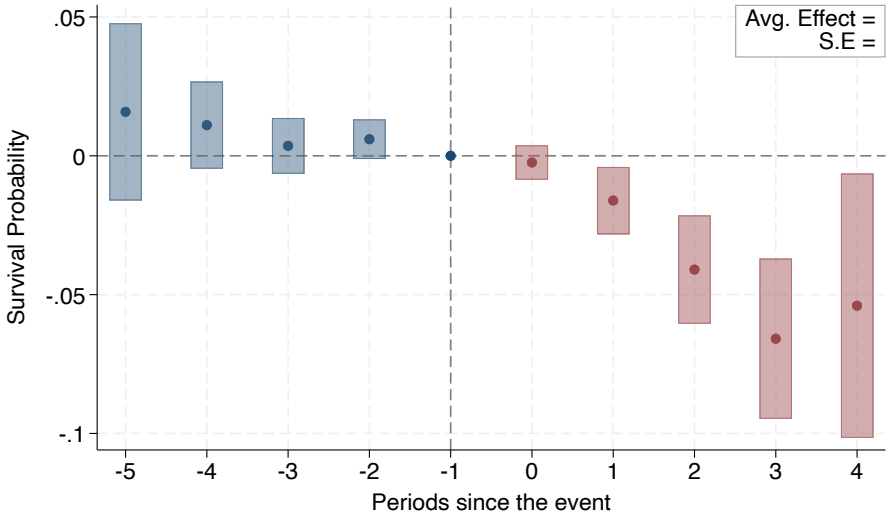
Robustness at t and $t+1$

- Redefine exit as the absence of a corporate income tax (CIT) return for at least two consecutive years.
- Restrict exit to absence of CIT return in year t and no record in employee tax returns in both years t and $t+1$

(a) No CIT at t and $t+1$



(b) No CIT at t and No PAYE at t and $t+1$



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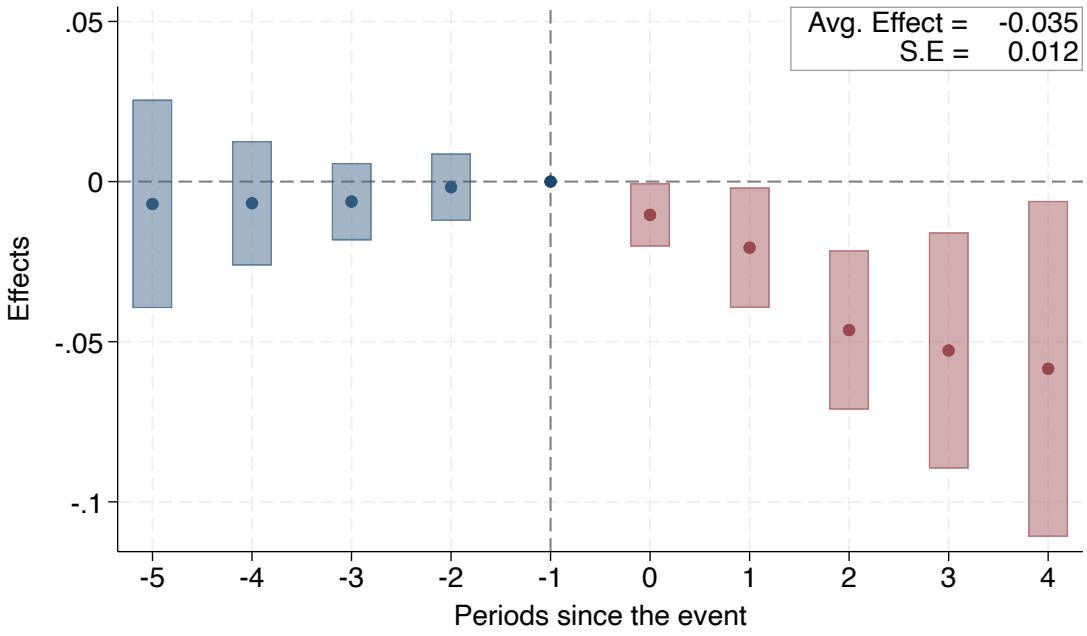
Survival Effects: Sample of Firms Present in First Year



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Limitation PIT Data

- "Due to the lack of TINs for a portion of employees, the PAYE datasets are only uniquely identified at the employer level and not at the employee level".
- "The variable c_tin_of_employee is for now not an interest of and not validated by the URA".
- "This results in many employees appearing to have the same TIN, no matter if the TIN is a 20-digit anonymized TIN from the real TIN or a 30-digit pseudo-TIN"
- "The suggestion here for researchers is to not use the variable c_tin_of_employee in analyses, because it is not validated by the URA and unreliable to be used. Therefore, we do not recommend using the datasets as employee-level panel".
- "However, each observation still pertains to unique tax liability filling for each employee working in the firms for a particular return month and year". Back